

Sahil Shah

Engineering Portfolio

Mechanical Engineering Graduate | University of North Carolina at Charlotte | May 2026
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ABOUT ME

I am a recent Mechanical Engineering graduate from UNC Charlotte with project experience in aerodynamic design, robotic mechanisms, and additive manufacturing. My work follows a consistent process: analyze the problem with simulation, build a prototype, test it against real data, and refine. I am comfortable owning a design from concept through fabrication and test, and I work well in multidisciplinary team environments.

TECHNICAL SKILLS

CAD and Modeling: SolidWorks, Creo Parametric, AutoCAD, Revit

Simulation and Analysis: MATLAB, MathCAD, Blade Element Momentum (BEM) analysis, aerodynamic simulation, QBlade, FEA

Manufacturing and Fabrication: FDM 3D printing, SLA resin printing, rapid prototyping

Programming: Python, C++, Arduino (sensor integration and control logic)

Software and Tools: Microsoft 365, Prusa Slicer, Arduino IDE, QBlade, MATLAB

PROFESSIONAL EXPERIENCE

Engineering Intern | M3 Engineering | May to Aug 2024

- Organized and consolidated technical documentation packages inherited from a prior engineering firm on an active industrial project, including engineering drawings, project records, and design deliverables.
- Reviewed drawing sets to establish document control structure aligned with M3's project standards and deliverable requirements.
- Collaborated with engineering staff to ensure documentation accuracy.

Teaching Assistant | Thermodynamics II | UNC Charlotte | Aug 2025 – May 2026

- Supported 80+ students in understanding complex thermodynamic principles through organized review sessions, study guides, and direct technical instruction.
- Evaluated design exercises and homework to provide constructive feedback, reinforcing problem-solving and analytical skills.

PROJECT 1: COLLEGIATE WIND COMPETITION 2026: WIND TURBINE BLADE DESIGN

Role: Blade Design, Manufacturing, and Testing Lead (Mechanical Sub-Team)

Team: UNCC Wind Design, 8 members across Mechanical and Electrical sub-teams

Project Overview

Our team designed, built, and tested a small-scale wind turbine for the 2026 DOE Collegiate Wind Competition. The goal was to produce as much power as possible across 5 to 10 m/s wind speeds while staying within a 0.45 m rotor diameter limit and surviving 13 m/s survival conditions. The turbine was scored across four tasks: power curve, rated power, emergency stop, and load disconnect stop. My work covered all aspects of blade design, from airfoil selection through manufacturing and field testing.

Engineering Challenge

The rotor needed to spin up at low wind speeds, run efficiently across a wide speed range, and hold together at survival loads. Most parts had to be 3D printable given our budget. The university had no wind tunnel, so we also needed a way to generate real performance data without one.

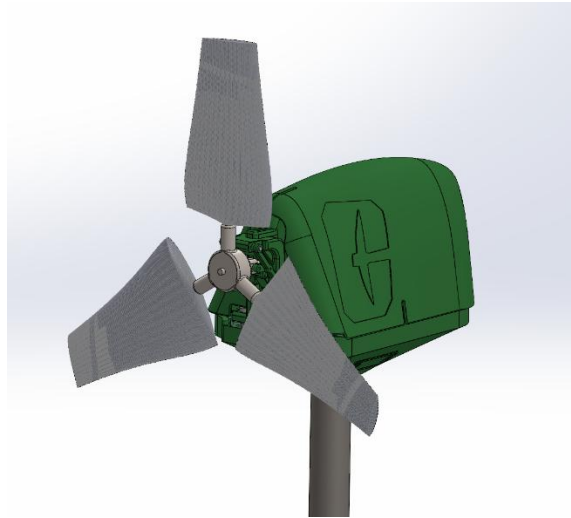


Figure 1.1: CAD rendering of the final UNCC Wind Design turbine

Airfoil Selection

The rotor diameter limit placed blade Reynolds numbers between 40,000 and 140,000. We evaluated S822, S823, SD7003, SD7062, SG6042, SG6043, MH112, NACA2414, S122, BW-3, Davis 3-R, and A18 airfoils through simulation and fabricated blades with S822, SD7003, Davis 3-R, and A18 airfoils. Each was simulated in QBlade and tested physically on the vehicle rig.

The S822 airfoil was my final choice as thinner profiles could not cut in reliably below 7 m/s. The S822 produced a higher lift-to-drag ratio across a wider angle-of-attack range, which gave it better startup behavior and more consistent output at the low wind speeds that carry the most weight in scoring.

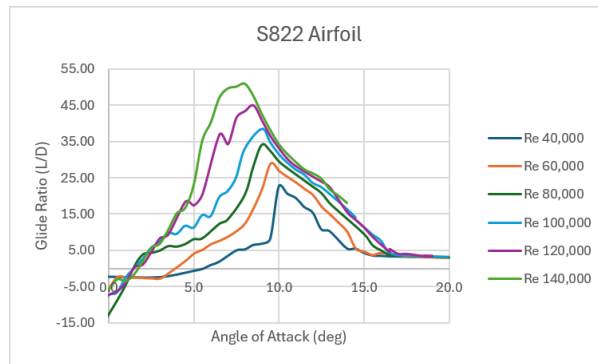


Figure 1.2: S822 glide ratio (L/D) vs. angle of attack across the operating Reynolds number range

Blade Geometry

I optimized blade twist and chord distribution using Blade Element Momentum (BEM) theory with the Schmitz optimization method in QBlade 2.0.9. The method sets each blade section to operate at its local optimal angle of attack, balancing lift forces along the span.

I ran iterative simulations to pick a design tip speed ratio (TSR). A TSR of 3.3 gave the best balance between early cut-in and peak power coefficient. Lower values helped startup but cut peak efficiency; higher values pushed peak efficiency higher but made cut-in worse and lowered output at low speeds.

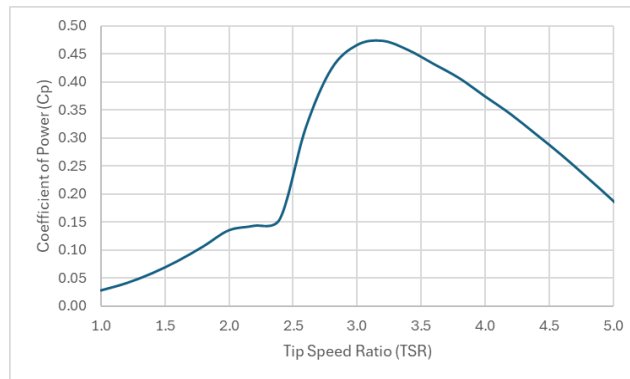


Figure 1.3: Simulated power coefficient (C_p) vs. tip speed ratio for the S822-O3.3 blade design

Manufacturing

Blades were printed in PLA+ using FDM. I designed a variable internal structure to get the right strength-to-weight balance across the span:

- Root (first 3 cm): 2.8 mm shell, 15% infill. Sized to handle bending moments from aerodynamic loading.
- Rest of the Blade: hollow with a 1.2 mm shell. Keeps rotational inertia and centrifugal loading low.

Structural bending tests on the S822 blade showed failure at 27.2 N with 9 degrees of deflection, giving a safety factor of 2.47 against the maximum expected aerodynamic thrust of 11 N. The S822 profile is thick enough that PLA+ did not need reinforcement.

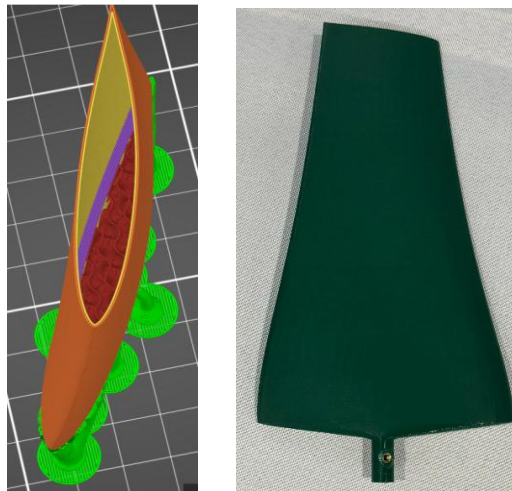


Figure 1.4: S822-O3.3 final blade design (left: PrusaSlicer; right: printed blade)

Testing

Without a wind tunnel, I designed a vehicle-based test method. The turbine was mounted on top of a car and driven at speeds matching the 5 to 10 m/s target range. A pitot tube on the tower and a handheld anemometer outside the car gave independent wind speed readings at the rotor. We ran multiple passes at each speed and averaged the results to reduce the effect of turbulence and unsteady flow.

Fan testing in the lab covered early-stage sensor calibration, pitch system checks, and e-stop verification. Vehicle testing was used for the blade performance data that determined the final blade selection.



Figure 1.5: Vehicle test setup (left: turbine mounted to vehicle; right: pitot tube and instrumentation)

Results

The S822-O3.3 blade outperformed all competing designs. Measured performance data from vehicle testing:

Wind Speed (m/s)	RPM	Power (W)
5	690	4.9
6	916	7.6
7	1,059	11.6
8	1,181	14.8
9	1,240	18.1
10	1,441	26.1

Measured TSR across the range was 2.9 to 3.5, which matched the simulated design value of 3.3 closely. This confirmed that the BEM optimization transferred accurately to the physical blade. C_p ranged from 0.22 to 0.43. Peak rated power was 26.1 W at 1,441 RPM.

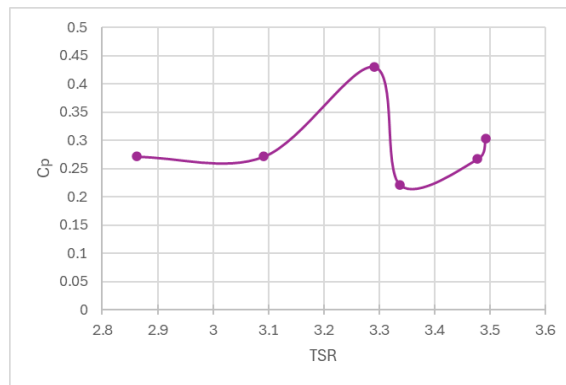


Figure 1.6: Measured C_p vs. TSR from vehicle testing; compare to Figure 1.3 for simulation agreement

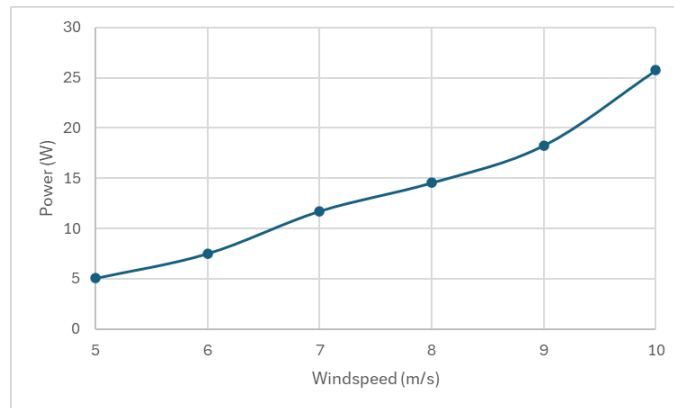


Figure 1.7: Measured power output vs. wind speed for the S822-O3.3 rotor

PROJECT 2: ROBOTIC PICK-AND-PLACE SYSTEM / GRIPPER DESIGN

Role: Gripper Design, Fabrication, and Integration

Team: 7 Mechanical Engineering majors.

Project Overview

The team built a 3-DOF robotic pick-and-place arm to sort and stack wooden cubes by color. My scope was the gripper: all design generations, force and torque analysis, CAD models, dimensioned drawings, fabrication, and integration with the third robot arm link (RA3).

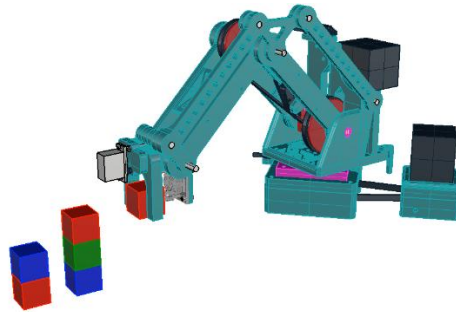


Figure 2.1: Full robot assembly CAD rendering; gripper is at the end of RA3

Design History

Generation 1: Rack-and-Pinion Tong Gripper

The first concept used two gear tongs pivoting on a shared axis, driven by a servo motor through a double-sided rack. Before building anything, I ran force and torque calculations to confirm the servo could hold a block during arm motion:

- Required gripping force: ~ 0.832 N rough calculation
- Required servo torque from geometry: ~ 0.068 Nm at the rack, minimum 1 kg-cm servo.

The rack-and-pinion added complexity without a meaningful benefit over direct drive, so it was cut in the next iteration.

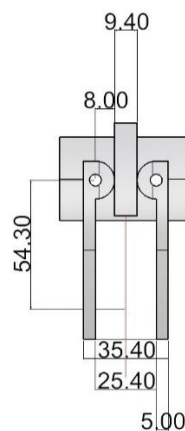


Figure 2.2: Generation 1 gripper with dimensions (mm)

Generation 2: Direct-Drive Servo Tong Gripper

The rack was removed. The MG90s servo horn was connected directly to the driving tong. The tongs are geared to each other at the pivot so that actuating one opens or closes both symmetrically. A TCS3200 color sensor was mounted directly behind the tongs to read the cube color at the moment of pickup.

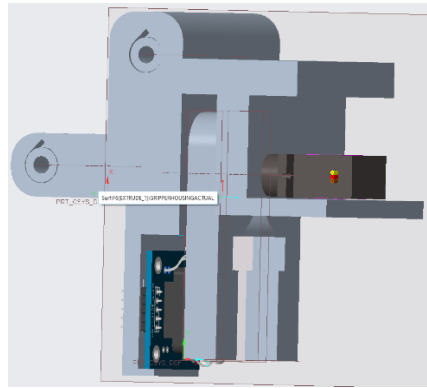


Figure 2.3: Generation 2 gripper with direct-drive servo and integrated color sensor

Generation 3: Final Design

The final iteration redistributed material to reduce mass and opened up clearance between the gripper housing and the color sensor to eliminate interference that appeared in testing. The gripper connects to RA3 through two 5 mm steel shafts running on deep-groove ball bearings. A rocker arm keeps the gripper level as the arm moves through its range.

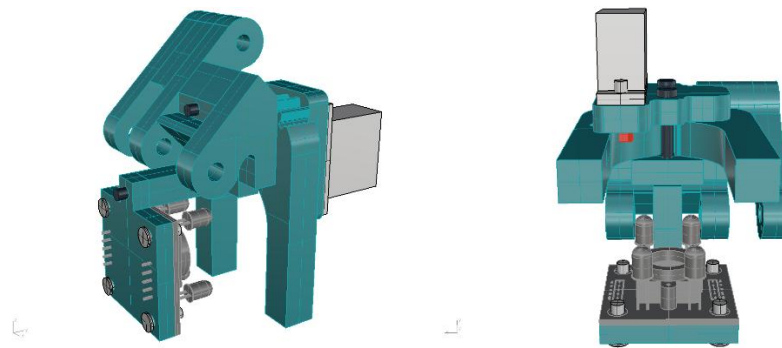


Figure 2.4: Generation 3 gripper with direct-drive servo and integrated color sensor

Torque Analysis for Final Design

I recalculated force and torque requirements using the final geometry and updated block mass:

- Required gripping force: 0.507 N
- Required motor torque: 0.21 kg-cm
- Motor selected: MG90s Micro Servo at 1.8 kg-cm rated torque; safety factor = 8.57

Total gripper mass including all hardware and electronics: 117 g.

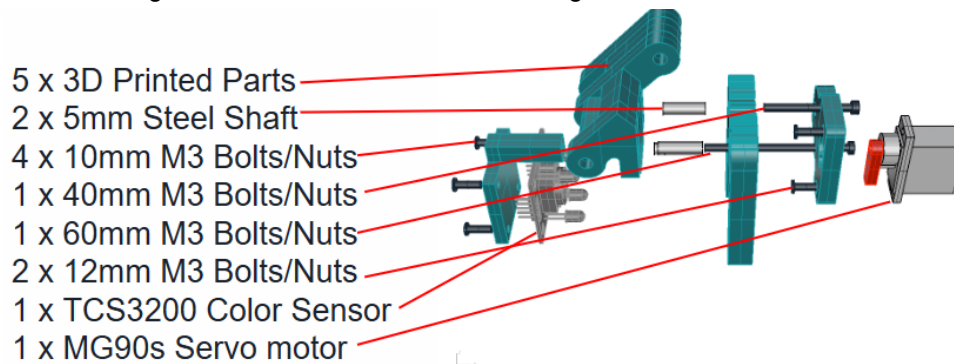


Figure 2.5: Exploded view of final gripper assembly

Drawings

Formal engineering drawings were produced for all five printed parts to UNCC drawing standards, including tolerances, scale, and title block information.

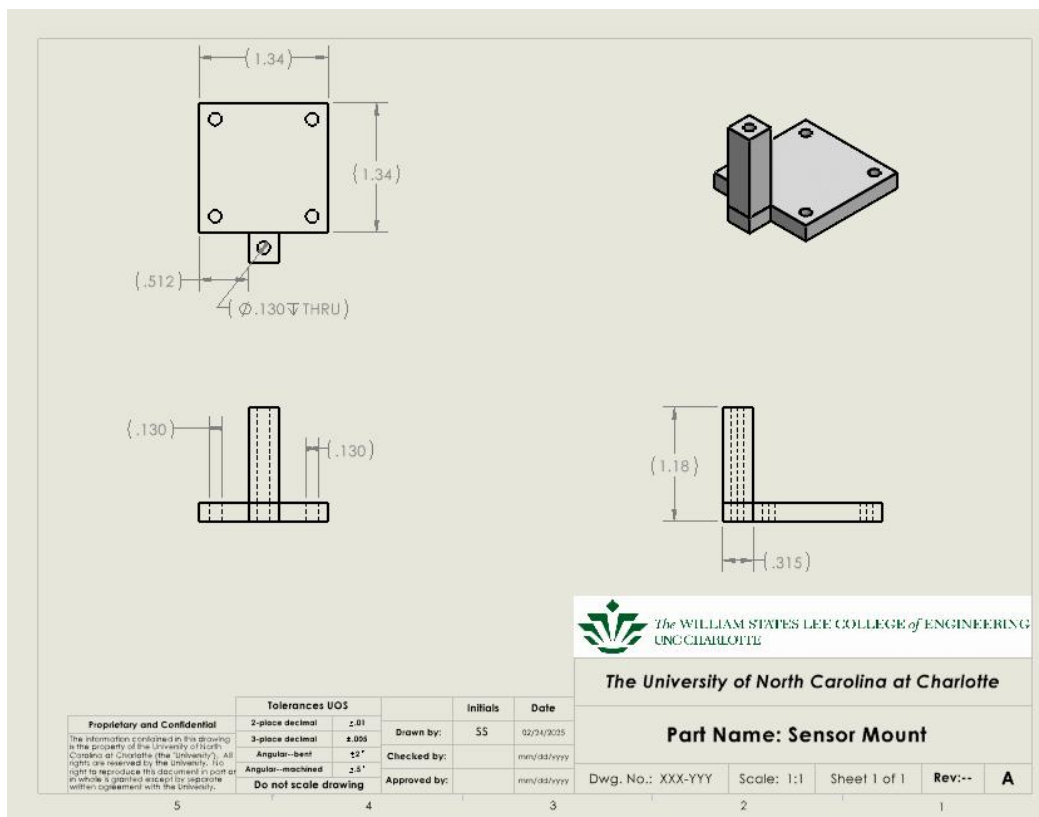


Figure 2.6: Dimensioned drawing, Sensor Mount

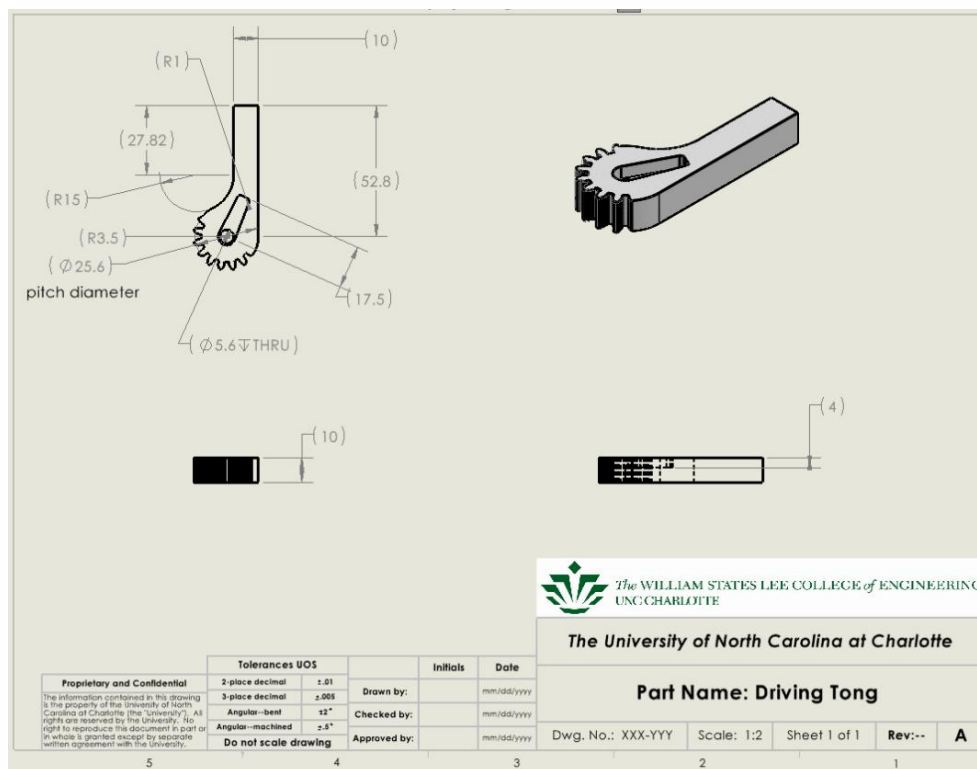


Figure 2.7: Dimensioned drawing, Driving Tong

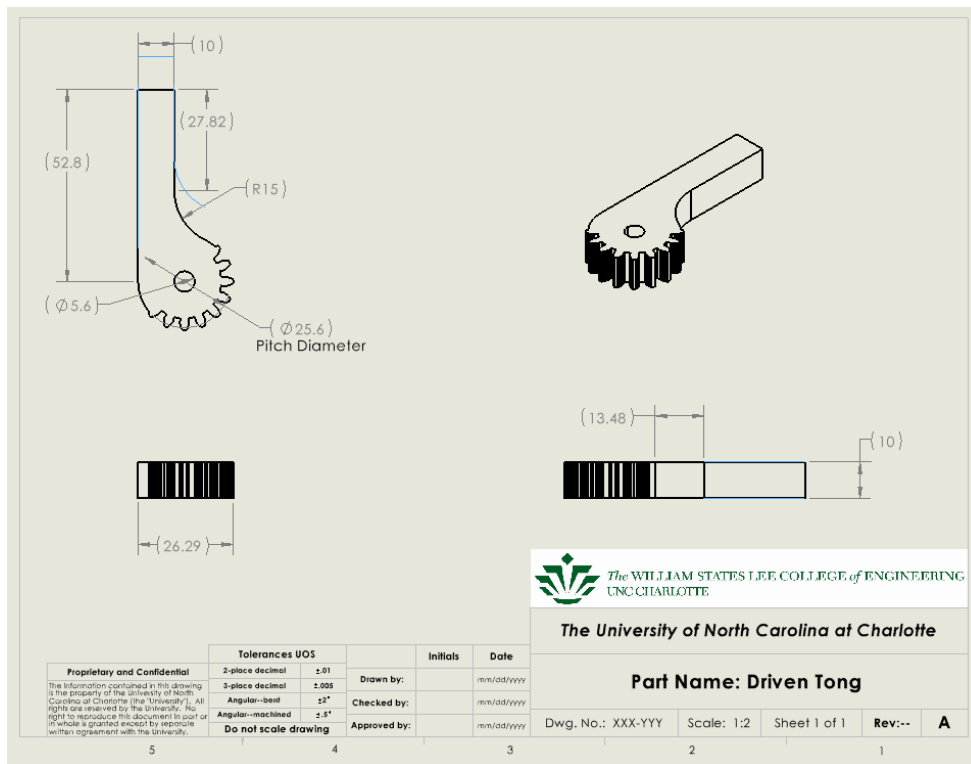


Figure 2.8: Dimensioned drawing, Driven Tong

Results

The robot picked up, sorted, and stacked all 9 cubes by color in 1 minute and 40 seconds. The gripper ran through repeated cycles without part failure, sensor error, or binding.

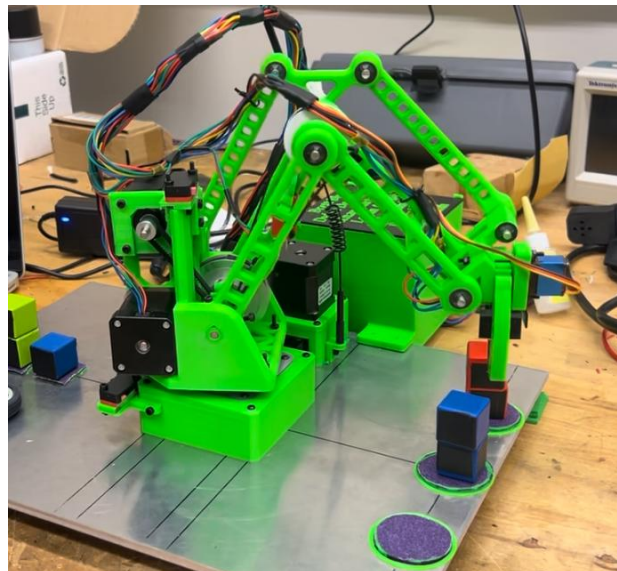


Figure 2.9: Robot during operation; cubes sorted and stacked by color

ENGINEERING APPROACH

Both projects followed the same process: start with analysis to narrow the design space, build a prototype quickly using 3D printing, test with real instrumentation, and use the data to decide whether to move forward or iterate. I find this cycle works well for problems where the theoretical model and the physical behavior do not always agree, and where testing is cheap enough to run frequently.

I am comfortable working across the full design process, from early concept analysis through drawing production, fabrication, and final test. I have also worked in team settings where my subsystem had to interface with other group's hardware and software, which required clear documentation and consistent communication throughout the project.